Rishikesh Kumar

ML Engineer

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Education

IIT Bombay | *M.Tech in Industrial Engineering and Operations Research (IEOR)* **Jul 2024 – Jul 2026**

Guru Ghasidas University | *B.Tech in Computer Science and Engineering (CSE)* **Jul 2020 – Jul 2024**

- **Relevant Coursework**: Deep Learning, Reinforcement Learning, NLP, Generative & Agentic AI, ML Principles, Operations Research

Internships

Full-Stack Developer Intern | *Vajra Labs, IIT Bombay* | *Powai, Maharashtra* **Aug 2025 – Dec 2025**

- Built a **malware analysis platform** with automated **feature extraction pipelines** for static, dynamic, and memory analysis
- Engineered **sandboxed multi-VM execution** with an **asynchronous Celery pipeline** for high-throughput concurrent analysis
- Led a **3-member team** using **Agile sprints**, delivering across **2 concurrent projects** on schedule

Projects

Explainable Depression Detection from Social Media | *Course Project* | *Prof. Pushpak Bhattacharyya* **Aug 2025 – Nov 2025**

- Built an **NLP pipeline for depression detection** from social media posts, using **double domain adaptation** and **guided masking**
- Designed an **explainability pipeline** by combining **Integrated Gradients** with an **LLM-based teacher–student distillation method**
- Achieved an **F1 score of 0.78** on test data, and extracted top-k words contributing most strongly to positive depression detection

The Third Eye: VIntel Platform | *Multi-Agent LLM Cybersecurity Intelligence System* **Jan 2026 – May 2026**

- Designed a **multi-agent LLM pipeline** with two-round editorial deliberation and **Borda-count** cross-story ranking algorithm
- Built **persistent memory** and **A2A protocol** for autonomous agent coordination across polyglot microservices (Go, TS, Rust)

HeadBall2 Game Player | *Self Project* | *Real-Time Game Automation with CV and RL* **Feb 2025 – Mar 2025**

- Developed a **real-time computer vision** pipeline for mobile game automation using **OpenCV**, from live video streams at **30 FPS**
- Designed a **reinforcement learning system** with **WebSocket-based** client-server communication, achieving **50 ms** end-to-end latency

Research & Publication

Classification of Plant Diseases using SOTA Deep learning Algorithms **July 2024**

International Journal of Computing | ***DOI:*** *10.47839/ijc.23.4.3542*

- Implemented and evaluated multiple **CNN architectures** on a dataset of **87,900+ leaf images** across **38 classes**
- Achieved up to **97.31% validation accuracy**, showing robust generalization across multiple plant diseases

MTP Thesis & Seminar

Parameter Specific Literature Retrieval | *Master's Thesis Project (MTP)* | *Prof. Balamurugan Palaniappan* **Jul 2025 – Nov 2025**

- Streamlined a **biomedical literature retrieval pipeline** to search PubMed data and extract **parameter-specific** clinical evidence
- Improved **entity extraction F1-score to 90%** by fine-tuning a **PubMedBERT-based NER model** on a manually annotated dataset
- Converted unstructured NER output into structured XML using a **BART-based Seq2Tree model** for parameter range extraction
- Implemented a **distributed ML pipeline** with **Ray, PyTorch, and HuggingFace Transformers**; deployed it with **Next.js/FastAPI**
- Built a **provenance-tracked Neo4j knowledge graph** and **QA engine** reaching an **88% answer rate** with every claim cited

Multimodal Transformer Architecture and Optimization | *Seminar* | *Prof. Urban Larsson* **Jan 2025 – Apr 2025**

- Implemented the **Attention is All You Need** paper in PyTorch and presented an in-depth analysis of the paper during the seminar
- Analyzed in-depth concepts of **An Image is Worth 16x16 Words** and presented how to use images in **transformer architecture**
- Conducted a seminar presentation on the research behind the **LoRA** paper and discussed its advantages and disadvantages

Technical Skills

- **Languages (Proficient)**: Python, C, C++, JavaScript, TypeScript, SQL **(Familiar)**: Go, Rust, Java, Bash
- **ML & Data**: PyTorch, HuggingFace Transformers, Scikit-Learn, NumPy, Pandas, Matplotlib, OpenCV, RLHF, LLM Agents
- **Frameworks**: Django, DRF, FastAPI, React.js, Next.js, Node.js, Tailwind CSS, Jupyter, Streamlit
- **Systems & Tools**: PostgreSQL, Neo4j, Redis, Docker, Docker Compose, Git, GitHub Actions, Ray, Celery, RabbitMQ, Linux

Position of Responsibility

Teaching Assistant | *IE 643, Deep Learning – Theory and Practice* **Jul 2025 – Nov 2025**

- Led **weekly programming sessions** for **80+ students**; managed grading, project planning, and regular doubt sessions

Achievements

- **Published** research paper on **Plant Disease Classification** using Deep Learning in **International Journal of Computing**
- Achieved **99.30 percentile (2024) and 99.13 percentile (2023)** in **GATE CSE** among 200,000+ candidates
- Secured **World Rank 32** in **CodeChef Starters 42** competitive programming contest
- Secured **Rank 1** in **HackForCG Hackathon** by developing an **ML-based computer vision** solution using **OpenCV** and **Haar Cascade**